

Visvesvaraya Technological University, Belagavi – 590018.



PROJECT PHASE-II REPORT
ON

Automated Hydroponic System

Submitted in partial fulfillment for the award of degree of

BACHELOR OF ENGINEERING
in
ELECTRONICS AND COMMUNICATION ENGINEERING

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**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING
CANARA ENGINEERING COLLEGE**

(An Autonomous Institution, Under VTU, Belagavi and Recognized by
AICTE, Accredited by NBA (CSE, ISE, ECE) and NAAC 'A' GRADE)

Sudhindra Nagar, Benjanapadavu, Mangaluru - 574219,

Karnataka.

2025-26

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CERTIFICATE

Certified that the project work entitled “**Automated Hydroponic System**” carried out by

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DECLARATION

We hereby declare that the entire work embodied in this Project Phase-II Report titled “**Automated Hydroponic System**” has been carried out by us at CANARA ENGINEERING COLLEGE, Mangaluru under the supervision of **Mr. Varun S Karadesai**, for the partial fulfillment of **Bachelor of Engineering in ELECTRONICS AND COMMUNICATION ENGINEERING**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Farming practices' own switch to meet the increasing demands for renewable food development and cautious use of natural resources. This function will enable the design of an IoT-enabled aquaculture system with no soil for the development of Green Chilly and Green Amaranth plants. In order to continuously read green information, the intended framework shall employ ground moisture and temperature–humidity detectors relevant to the NodeMCU ESP32 microcontroller. These readings are transmitted using Wi-Fi to a mobile platform called Blynk which allows users to monitor the structure in real time. With sensor critique, it ensures a proper flow of water and a suitable humidity for plant roots within the apparatus.

Keywords Hydroponics, IoT, NodeMCU, Blynk, Smart Farming, Sensors.

Table of Contents

Acknowledgement	i
Abstract	ii
Table of Contents	v
1 Introduction	1
1.1 Background	1
1.2 Motivation and Problem statement	2
1.3 Objectives	2
1.4 Proposed Methodology	3
1.5 Scope and limitations	4
2 Literature Survey	5
2.1 Article 1: Production and Utilisation of Hydroponics Fod- der	5
2.1.1 Brief Findings	5
2.1.2 Design/Methodology/Techniques	5
2.1.3 Result Achieved	6
2.2 Article 2: Hydroponics	6
2.2.1 Brief Findings	6
2.2.2 Design/Methodology/Techniques	7
2.2.3 Result Achieved	7
2.3 Article 3: Environmental impacts of urban hydroponics in Europe	7
2.3.1 Brief Findings	7

2.3.2	Design/Methodology/Techniques	8
2.3.3	Result Achieved	8
2.4	Article 4: IRRIGATION CONTROL IN HYDROPONICS	9
2.4.1	Brief Findings	9
2.4.2	Design/Methodology/Techniques	9
2.4.3	Result Achieved	10
2.5	Article 5: HYDROPONICS	10
2.5.1	Brief Findings	10
2.5.2	Design/Methodology/Techniques	11
2.5.3	Result Achieved	11
2.6	Article 6: Crop Hydroponics, Phyto-hydroponics, Crop Pro- duction, and Factors Affecting Soilless Culture	12
2.6.1	Brief Findings	12
2.6.2	Design/Methodology/Techniques	12
2.6.3	Result Achieved	12
2.7	Article 7: Monitoring of Hydroponics System using IoT Technology	13
2.7.1	Brief Findings	13
2.7.2	Design/Methodology/Techniques	13
2.7.3	Result Achieved	14
2.8	Article 8: Optimization and Control of Hydroponics Agri- culture using IOT	14
2.8.1	Brief Findings	14
2.8.2	Design/Methodology/Techniques	15
2.8.3	Result Achieved	15
2.9	Gaps Identified and Addressed	16
2.10	Summary	16
3	Software Requirements Specification	18
3.1	Functional Requirements	18
3.2	Non-Functional Requirements	18
3.2.1	Safety Requirements	19
3.2.2	Performance Requirements	19

3.3	Hardware and Software Requirements	19
3.4	Summary	20
4	System Design	21
4.1	Abstract Design	21
4.1.1	Architectural diagram	21
4.2	Functional Design	22
4.2.1	Modular design diagram	22
4.2.2	Sequence diagram	23
4.2.3	Use case diagram	25
4.3	Control Flow Design	25
4.3.1	Activity diagram for use cases	25
4.4	Summary	25
5	Implementation	27
5.1	Hardware Implementation	27
5.2	Software Implementation	27
5.3	Summary	28
6	Conclusions and Future Work	29
	References	30
A	Drill-bit	30
B	Project-Expo Details	31

Chapter 1

Introduction

1.1 Background

Hydroponics is a current method of growing plants without using soil. Alternatively, plants get their nutrition from water which is mixed with nutrients. Even though it seems to prefer a fresh idea, citizens have been growing plants free of ground for centuries. For instance, the Babylonians used to have the famous hanging gardens, and the Aztecs turned their plants into floating gardens. During the 19th and 20th centuries, scientists took hydroponics seriously. They found that plants wearing thymine actually need land for their growth, just water, essential elements, sunlight, and gas. Earth's all that's left of a plant in a location. The present idea has led to new development processes using components such as coconut peat, perlite, clay ball, rockwool, and even systems using just water. Hydroponics, as long as it enabled food to be produced on topographic points alongside impoverished or very small areas, became more fashionable in later Universe battle II. Nowadays, hydroponics is used by citizens in towns, greenhouses, rooftops, vertical farms, and large commercial farms. The current technology allows plants to grow faster, produce more, and last throughout the year while using a lot less water than conventional farming. Currently, hydroponics is considered to be an essential farming machine for the future. It helps decipher food problems, saves space, and allows farms even in tough environments to admire cities in alternately dry regions.

1.2 Motivation and Problem statement

There is a huge gap between the development of food and the space and resources we possess in today's second world. Multiple individuals live in towns where soil is restricted, earth's quality is poor, and the climate is unpredictable. The present situation makes farming difficult. Hydroponics motivates one to think about the fact that we can grow a fresh, healthy plant that does not depend on Earth, using less water, less space, and less effort. In order to facilitate the approach to farming, it feels that it prefers a smart and green way. I'd like to study and contribute to the current sanitary and simplified plant growth strategy. The classic agricultural practice faces many challenges—land pollution, insufficiency, water shortage, plague, and reliance on the weather.

1.3 Objectives

- **Observing and Recording Plant Growth:**

To normally proctor plant height, color, and general health and to determine which conditions facilitate plant growth more rapidly and more rapidly.

- **Maintaining the System Easily:**

To find simple, low-cost methods to support aquaculture equipment so that anyone, students or novices, could use it without difficulty.

- **Understanding System Challenges:**

To learn the common challenges that can arise in the aquaculture system, such as incorrect food levels, decreased water levels, alga growth, or pump problems, and to figure out how to repair them so that the plant remains robust and grows properly.

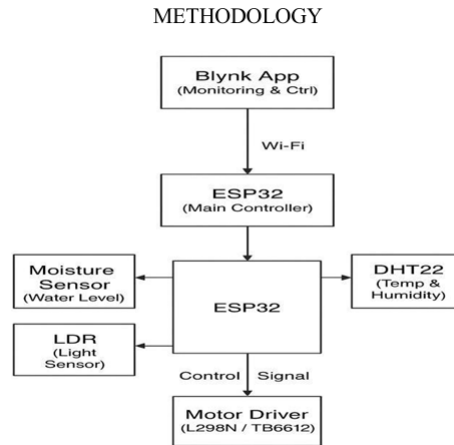


Figure 1.1: Proposed Methodology

1.4 Proposed Methodology

The method of the current hydroponics venture consists in choosing simple, aggressive plants such as boodle, spinach, or a batch, which is ideal for novices and helps understand the way aquaculture systems work. After selecting the plant, the next step is to establish the basic hydroponic system. The apparatus comprises a container or tray for holding the water and a correct growing medium, i.e. coconut sand, perlite, or a clay ball in the back of the plant, and a feeding solution that provides the plant with the essential mineral it normally obtains from the ground. The nutrient solution shall be composed of purified water and aquiculture feed in an appropriate proportion. Once the system is ready, young saplings are gently transplanted into the growing medium to ensure their roots touch the nutrient-rich water. That allows the plant to be able to absorb the essential elements immediately. Appropriate lighting shall be provided either by natural sunlight or by means of a developing light for the next to the lowest 6–8 times a day. In addition, proper ventilation and air circulation are necessary to maintain the plant's good health and prevent the development of mold. Regular monitoring shall be transferred outside during the course of the activity. Periodically check the water stage and replenish it if necessary. The feeding concentration must be measured and adjusted to maintain a balanced level since plants in hydroponics depend entirely on

this solution.

1.5 Scope and limitations

Scope

The project also involves monitoring water levels, food balance, and plant health to see how they compete in a controlled setting.

Limitations

The Main limitations of hydroponic is mainly it requires very stable maintainance , it requires very high cost for maintaining the setup.

Chapter 2

Literature Survey

2.1 Article 1: Production and Utilisation of Hydroponics Fodder

2.1.1 Brief Findings

Hydroponics companies have shown that plants can grow skillfully even without soil, provided the right mixture of water and nutrients are provided. Compared to soil-grown plants, they grow faster and appear more healthy compared to their roots. The framework applied a great deal less water and the plant prerequisites less space, which was acceptable to small sectors which appreciated the balcony. Nevertheless, the project has shown that hydroponics requires regular monitoring of the water temperature, nutrition, and pH in order to maintain the robustness of the plant. Generally speaking, the discoveries suggest that hydroponics is an efficient and space-saving method of growing plants, particularly in topographic regions where land or topography is restricted. .

2.1.2 Design/Methodology/Techniques

The hydroponics system is designed to support plant growth that does not have soil by providing the roots with a pathway into the water and nutrients.

2.1.3 Result Achieved

By integrating Internet of Things (IoT) technology into aquaculture farming arrangements, the modern agricultural blockade can be solved with high productivity and resilience. The results of this study show that real-time monitoring of key parameters by IoT-based aquicultural frameworks can maximize crop development and pruning waste. IoT frameworks ensure that the ideal growing conditions are maintained, and their consequences are better plants and higher output by providing real-time information on important parameters such as temperature, humidity, pH, and food degrees.

2.2 Article 2: Hydroponics

2.2.1 Brief Findings

The authors explain that hydroponics, growing plants in nutrient-rich water instead of soil, is becoming an increasingly important farming technique. It stresses that a number of separate expanding elements prefer clay ball, coconut sand, perlite, and even wood fiber to support a plant free of soil. Hydroponics has been described to remain able to grow vegetables such as tomato, boodle, cucumber, pepper, as well as cosmetic plants such as rose, freesia, and herbs. Traditionally, the theory has existed for a decade with Dr. Gericke first popularized the term in 1929 in the United States and Canada. The army was able to use hydroponics during Universe Battle II to produce fresh food on an island that had no fertile soil. The conclusions indicate that hydroponics enables farming in areas where land farming is not feasible, provides better management than essential elements, saves water, accelerates plant development, reduces plague and disease, increases crop production, and reduces labor costs equivalent to weeds. Generally speaking, the paper suggests that hydroponics is a realistic, efficient, and high yielding system for modern farming.

2.2.2 Design/Methodology/Techniques

The hydroponics framework in the current text is formulated as the idea of growing plants without soil using a nutrient-rich water solution and, assuming necessity, lightweight development components such as clay, coir, perlite, or vermiculite. The basic technique involves planting the roots of the plant directly in the current nutrient solution so that they are able to easily absorb all that they need to enhance. The system imposes packed management beyond food, pH, and sustainable conditions, enabling plants to develop faster and more healthily. The structure requires the establishment of a balanced diet, the maintenance of oxygenated water, and the constant monitoring of the elements which enjoy temperature and sourness. The plants are supported by containers or channels, and water is either continuously flowing or supplied within a given time interval. Given that there is no land there, all food is supplied by means of water, making monitoring an obligatory technique. The structure reduces plague, addition yield, and efficiency even in topographic areas where conventional soil-based farming is not used for academic writing.

2.2.3 Result Achieved

The results of hydroponics have been very helpful. we got the thigspeek results in terms of the graph which shows the humidity level and the moisture level of the hydroponic system

2.3 Article 3: Environmental impacts of urban hydroponics in Europe

2.3.1 Brief Findings

The study found that urban hydroponics could be a promising technique for expanding food supply in cities, despite the fact that its ecological change depends entirely on the design of the organization. The aquaculture industry in Lyon is showing its support for streamlining water usage

and the ability to grow fresh vegetables close to the consumer, thereby reducing pollution from transport. However, the study also shows that hydroponics can use a large amount of energy, particularly light, heating, and structure management, in addition to indicating outside that hydroponics can be used. In this way, the overall sustainable footprint can reach a very high level, assuming that energy sources are renewable. Furthermore, the study found that the environmental impact of using renewable energy or sustainable equipment is significantly reduced, making hydroponics a lot more green. Overall, the conclusions suggest that hydroponics can remain sustainably friendly, but they are produced merely to provide intelligent options roundabout energy application, substances, and structural control.

2.3.2 Design/Methodology/Techniques

In this study, the researchers examine the way the urban aquaculture farm in Lyon affects the surroundings. Instead of simply thinking, they used a detailed empirical strategy called Life Cycle Assessment (LCA). This tactic seems to be close to everything that travels through the hydroponic system, from the components used to build it, the energy needed to operate it, the water and feed used, and even the way the produce is finally transported to the clients. They gathered real facts from an aquaculture farm, such as electricity usage, light duration, heating requirements, and food consumption. They also want to see which one has a smaller sustainable footprint, in addition to comparing such numbers to conventional farming. The partnership used a computer model to calculate how each stage construction, growth, gathering, packaging, and distribution affects the environment in terms of carbon emissions, energy consumption, water usage, and waste coefficients.

2.3.3 Result Achieved

Research shows that urban hydroponics can be both helpful and destructive to the atmosphere. On the positive side, the hydroponic system uses a lot less water than traditional soil farming and produces fresh vegetables

right in the city, thus reducing the pollution of the transport system. It also allows year-round production and reliable food equipment even during periods when land farming suffers. However, the consequences have also revealed a major problem with energy use. The aquaculture farm depended entirely on, on top of electricity to light, pump, and temperature control. The carbon footprint of the framework compared with that of the expected was lower as long as the energy comes from non-renewable sources. The increased energy demand partially eliminates the environmental benefits of the present method. Overall, the argument is that hydroponics can remain environmentally friendly, despite the fact that the structure relies solely on the use of sustainable equipment or alternatively renewable controls, beginning to appreciate the solar resource. At the same time, hydroponics is developing into a greener and more environmentally friendly option to feed the city.

2.4 Article 4: IRRIGATION CONTROL IN HYDROPONICS

2.4.1 Brief Findings

The researchers have shown that urban hydroponics can remain together, be helpful and difficult to cope with the conditions. which we have found that the hydroponic is very much essential and provides an easier growth of the plants in a very sequential way

2.4.2 Design/Methodology/Techniques

The report states that an aquaculture framework is planned to promote plant growth without soil by exposing its roots to a nutrient-rich water solution. it also provides very consistent growth to the plants in a very helpful way

2.4.3 Result Achieved

The results of the hydroponics study show that plants respond very competently when provided with the correct mixture of essential elements directly from water. Compared to conventional soil growing methods, hydroponic plants grow faster, have stronger roots, and produce a more constant increase compared to soil growing methods. Since the situation was controlled, the plant had precisely what it needed, no more or less, to improve quality and increase output. In addition to the creation of hydroponics, research aims at reducing the many common problems encountered on Earth, such as plague, disease, and food loss. Water and nourishment have been used more adequately, so the structure recycles them instead of draining them. The roots are clean, and the plant is quick to adjust its feeding solution. Overall, the results show that aquaculture plants can produce strong, healthy plants in a reliable and simplified method, proving that this tactic is a practical alternative to land farming.

2.5 Article 5: HYDROPONICS

2.5.1 Brief Findings

The paper underlines the use of hydroponics as a productive method to increase plants without soil by feeding them directly with water. One of the essential discoveries is that plants grown in aquaculture systems typically exhibit faster growth, better root health, and a more uniform evolution compared to soil plants. This is due to the plants accepting balanced nutrition, adequate oxygen, and a monitored environment. The sheet also shows that hydroponics helps avoid several soil-related problems such as plague, disease, and nutrition imbalance. The result is that plants are more robust and require less chemical treatment. Water should be plentiful but not excessively so since the organization recycles water instead of losing it admirably in soil farming.

2.5.2 Design/Methodology/Techniques

The report describes hydroponics using a simple design, where plants are grown without ground and, alternatively, their essential elements are obtained through water. A container for holding the feed solution, a support system for holding the clay pellets or other material in order to keep the plant upright, and a system for supplying oxygen to the roots. The aim of the design is to ensure that certain plant roots always have access to water, nourishment, and air. The method starts with the introduction of a balanced feeding solution containing all the essential minerals necessary for plant growth. Then the plant is placed in the structure so that its roots can reach the solution directly. During the procedure, the fluid must remain in motion or otherwise air escapes so that the roots of wear'thymine suffocate. The sheet also describes the monitoring elements such as pH, temperature, and food content to ensure the plant remains healthy. Adjusting the composition of the diet as the plant grows, checking oxygen levels in the root zone, and detecting plant responses to changes in the environment are key approaches that are familiar with. The organization creates an ideal growing environment where the plant can grow faster and more uniformly than in the ground, thanks to the control of these components. The above tactics support farmers to maintain good plant development while reducing the burden of plague or disease.

2.5.3 Result Achieved

The results from the hydroponics study showed that plants grown in water-based systems performed very well. they grew very much faster without any incovieneince and interruption in very smoother way which will also decreases the manual effort of the man

2.6 Article 6: Crop Hydroponics, Phyto-hydroponics, Crop Production, and Factors Affecting Soilless Culture

2.6.1 Brief Findings

The paper explains that hydroponics and phyto-hydroponics are effective ways to grow crops without soil by giving plants nutrients directly through water. The findings highlight that these soilless systems help plants grow faster, healthier, and more uniformly because they receive a balanced supply of water, nutrients, and oxygen. The study found that hydroponics can greatly improve crop production, especially in areas where soil quality is poor or land is limited.

Another important finding is that plant growth in hydroponics depends on several key factors such as pH, temperature, nutrient strength, light, and water quality. When these factors are controlled properly, the plants perform better than in traditional farming. The paper also points out that soilless culture reduces soil-borne diseases, saves water, and makes nutrient use more efficient.

2.6.2 Design/Methodology/Techniques

The paper describes hydroponics and phyto-hydroponics using a system designed to grow plants without soil by giving their roots direct access to nutrient-rich water.

By paying close attention to light, water quality, and nutrient balance, the techniques ensure that plants grow faster, stay healthier, and produce more than they typically would in soil.

2.6.3 Result Achieved

The results of the investigation show that plants grown in hydroponics and phyto-hydroponics have a lot more advantages than those grown in the ground. The plants moved faster, had stronger roots, and produced

more output because they took in nutrients directly from the water in the exact amounts they needed. The plant was subjected to less stress and a more uniform increase due to the fact that the organization did not allow for complete direct beyond the pH, temperature, and feeding stages. The analysis also shows that the use of water was very low because the feeding solution was recycled instead of being abandoned on the ground. The plants were faced with fewer challenges of plague and soil diseases, which helped them to stay healthy throughout their increasing duration. Generally speaking, the results show that soilless heritage is a productive, clean, and productive way of developing a crop, particularly in topographic regions where earth excellence is reduced or space is limited. .

2.7 Article 7: Monitoring of Hydroponics System using IoT Technology

2.7.1 Brief Findings

The article shows that IoT (Internet of Things) technology can make aquaculture farming much easier, smarter, and more reliable. The research shows that by adding a detector to the aquaculture framework, essential plant growth conditions such as temperature, humidity, water level, pH, and nutrition concentrations can be monitored instinctively in real time. The present eliminates the need for constant manual check. The results also show that the structure develops overly effective when the variables are monitored using IoT devices. Farmer or user gets instantaneous updates via mobile telephone to repair problems before plants are damaged. In order to increase the number of healthy plants, reduce the number of mistakes, and improve the overall quality of the crops, these guidelines are provided.

2.7.2 Design/Methodology/Techniques

The paper describes an aquiculture structure designed to work together with IoT technology for plant expansion, which is easy and additional spot.

2.7.3 Result Achieved

The investigation shows that adding IoT technology to a hydroponic organization has made plants much more efficient and easier to manage. The detectors were able to accurately measure essential elements such as pH, water content, temperature, humidity, and feeding stages, and the current information was sent immediately to the consumer via telephone or computer. The present presupposes that the structure can be monitored at any time and from any location. Problems such as minimal water levels or otherwise incorrect pH were quickly eliminated by real-time monitoring, thus preventing plant damage. The structure provided a perfect growing environment for plants, which led to healthier and faster-growing plants. It also reduces manual work since the consumer does not need to constantly check the organization. Generally speaking, the results indicate that IoT-based hydroponics provides a more intelligent, more reliable, and more productive farming framework. It improves accuracy, saves time, reduces errors, and ensures a higher quality of crop while reducing the number of attempts.

2.8 Article 8: Optimization and Control of Hydroponics Agriculture using IOT

2.8.1 Brief Findings

The analysis shows that the use of IoT equipment significantly enhances the mode of aquaculture production. The system can continuously monitor essential environments such as water, pH, temperature, humidity, and food concentrations by adding detectors and automation. The present invention makes the aquaculture structure more intelligent and more reliable, thereby reducing the need for manual supervision. The results show that the structure can immediately raise a buyer any time the product goes wrong, such as when the liquid becomes excessively acidic or the food layer falls. The structure can even spontaneously adapt to the situation,

allowing plants to grow under stable and healthy conditions.

2.8.2 Design/Methodology/Techniques

A system to automate an aquaculture farm using IoT is planned in the present document. They work by placing detectors at the aquiculture apparatus to continuously measure essential parameters such as water stage, pH, temperature, humidity, and nutrient tolerance. These detectors transmit the information to a microcontroller which then transmits the information to the internet so that the user can check it from his mobile phone. The method involves putting in place the initial hydroponic structure, putting the sensors in the correct position, and connecting them to an IoT board such as Arduino or NodeMCU. The sensed facts will be shown on the Internet splashboard, allowing the buyer to monitor the condition of the plant in real time. The system can also unconsciously turn on or off the pump, adjust the feed, or control the flow of water anchored on top of the detector reading. Unbroken detection, real-time data transmission, automation, and alert presentation are key strategies used. This makes aquaculture structures more precise and robust, reducing manual labor and ensuring that plants always accept the ideal development state.

2.8.3 Result Achieved

The results show that the IoT-based aquaculture farming system has a great advantage over the traditional manual monitoring. The organization could accurately measure all critical conditions and immediately alert the consumer to the need for attention. The present aided preventive problems include food imbalance, decreased water content, or alternatively incorrect pH. As the framework adjusts its conditions swiftly and without its conscious consent, the plant develops more robust and faster. Moreover, the analysis shows that IoT devices reduce human efforts, increase performance, and make aquiculture farming simple even for novices. Generally speaking, this structure provides better quality of crops, rescues the era,

and enhances the aquaculture apparatus, which is more solid and productive.

2.9 Gaps Identified and Addressed

Gaps Identified

- Requires pumps ,containers ,lights and sensore
- Must understand the ph ,nutrients ,levels etc
- If there is power cut it shoul work very easily

Gaps Addressed

- It is very redialy available
- It also provides very much need for the technical knowledgs
- Many systems now use battery backups or automatic controllers to protect plants during power cuts.
- To ensure the correct stability of the essential components, the electronic pH meter, the EC detector, and the ready-to-eat meal are easy to use.
- Institutes, colleges, and urban farms teach people about hydroponics and help others understand and adopt it.

2.10 Summary

Hydroponics is a modern way to grow plants that do not need soil by using nutrient-rich water instead. The current system enables plants to grow faster and more healthily, given that they obtain precisely what they need directly from the water??In our company, we are looking for ways to build a simple hydroponic structure, promote plant growth, and improve the state of light, water, and nutrients. We're also trying to find out what people enhance the top in this soilless environment by testing different types of

plants.??To show that hydroponics can be effective, durable, and suitable for topographic points together with a narrow terrain was one of the main objectives of our project. We are trying to automate and maintain the best growing conditions, making farming simpler and less labor-intensive by using detectors and simple controls.??Through this project, we have learned that hydroponics can produce additional crops with less space and time, use less water compared to conventional farming, and can be used even in towns and surrounding areas where the earth's excellence is underprivileged. Hydroponics is, in general, a smart and sustainable farming technique for the future.

Chapter 3

Software Requirements Specification

3.1 Functional Requirements

The operational conditions describe the compulsory procedures that the Blynk app must follow when it interacts with the aquaculture IoT framework.

- The app should be able to take readings from different sensors like temperature, water content.
- The app should be able to take readings from different sensors like temperature, water content.
- Whenever the sensors send data, the app should process it correctly and show it clearly.
- Whenever the sensors send data, the app should process it correctly and show it clearly.
- For every set of sensor readings, the app should provide accurate and timely results, like real-time graphs or alerts.

3.2 Non-Functional Requirements

The conditions given under non-functioning describe the manner in which the framework should work. It includes maintenance, reliability, serviceability, performance, and support.

3.2.1 Safety Requirements

- The system should run smoothly without crashing, even if it operates continuously for days.
- The system should be able to handle more sensors or expand to larger setups without problems.

3.2.2 Performance Requirements

- Even when multiple sensors are sending data simultaneously, the system must remain fast and efficient.
- It should be able to handle additional sensors or an expanded setup without affecting speed or accuracy.

3.3 Hardware and Software Requirements

Hardware Requirements

- ESP32 or NodeMCU microcontroller
- Sensors:
 - Moisture sensor
 - DHT22(Temperature and Humidity sensor)
- Water pump
- Relay module
- Power source (e.g., battery or adapter)

Software Requirements

- Blynk IoT Platform (Blynk 2.0)
- Arduino IDE (for writing and uploading firmware)
- Wi-Fi router

- Mobile device (Android/iOS) with the Blynk app installed

3.4 Summary

The Software Requirements Specification describes the way the AC Duct Cleaning Automaton uses the ESP32 microcontroller and ESP32-CAM. The software allows wireless control via a web interface, allowing the customer to move the automaton and view real-time video inside the conduit. It removes the motor direct, the functioning of the mechanism, the connection to the Internet, and the webcasting of the live webcam. The Software requirements Specification defines essential functional requirements, such as motion control, video transmission, command handling, and organizational feedback. In addition, the requirements for non-functioning requirements such as performance, reliability, serviceability, and safety are included to guarantee the smooth and safe functioning of the machine in confined spaces. Generally speaking, the Software Needs Specification provides a complete guideline to the creation, development, and testing of the software that can operate the duct cleaning machine.

Chapter 4

System Design

4.1 Abstract Design

4.1.1 Architectural diagram

The architecture diagram is supposed to give us a plan of how the structure was built in the manner in which it was built. The architecture diagram of our organization's construction is under it.



Figure 4.1: Architecture diagram

Hydroponics structure has a smart plant caretaker who has a unique position in all divisions. Sensors act as the body parts of the structure and continuously check vital environments such as pH, nutrient levels, temperature, humidity, and water level.

4.2 Functional Design

4.2.1 Modular design diagram

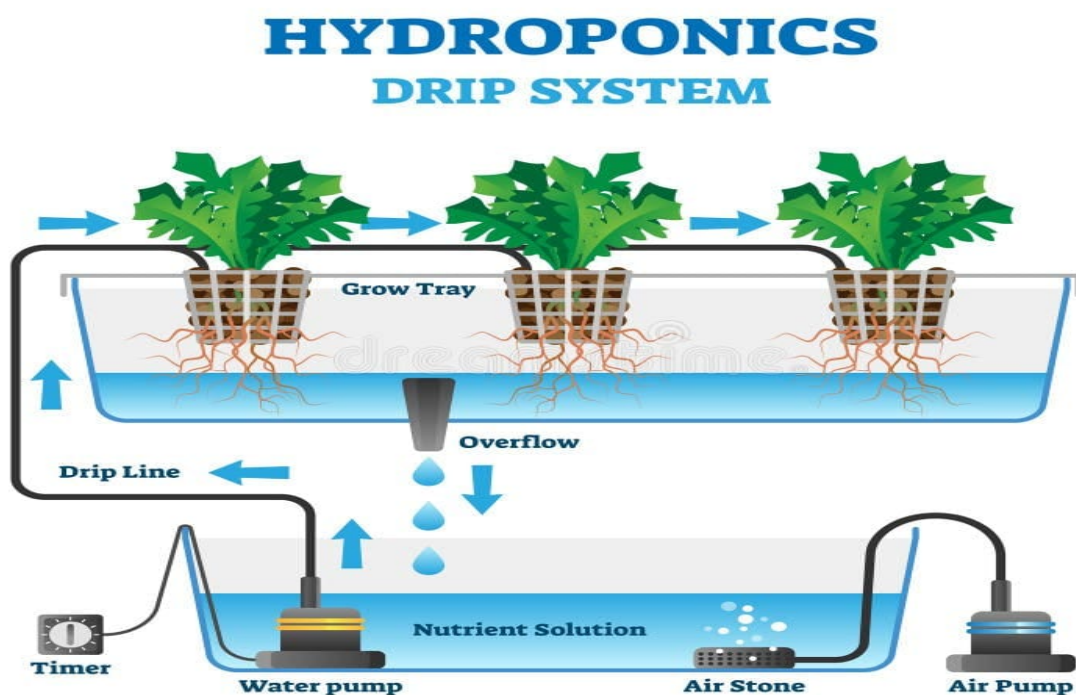


Figure 4.2: Modular design diagram

- Continue to monitor the similar context in the hydroponics system
- In this section the complete resolution is being made
- It is very helpful to trigger the actions of the setup
- It is used to transmit the data to the cloud

4.2.2 Sequence diagram

The consecutive diagram enables us to understand how an individual object interacts in a consecutive system.

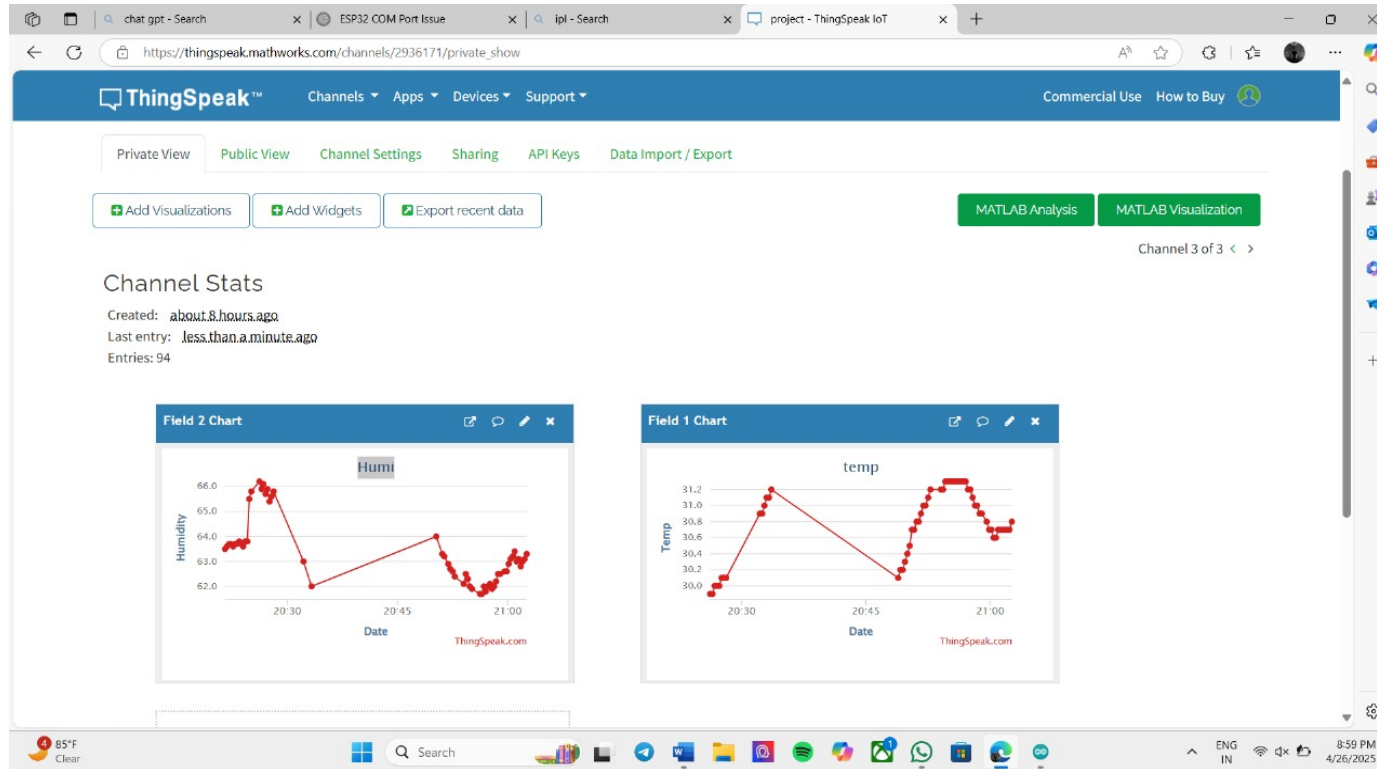


Figure 4.3: Sequence diagram

The sequence diagram of the hydroponics system shows how the parts are communicating with each other in a smooth manner, preferring a continuous dialogue. First, the detector checks the environment—measurements of pH, food, temperature, humidity, and water—and sends the current information to the accountant, who acts as the second brain. The accountant reads the statistics and decides who wants to do it next. If the object is out of symmetry, it sends commands to the actuators, such as the pump, the fan, or expanded light, specifying which ones to turn on or off to repair the object.

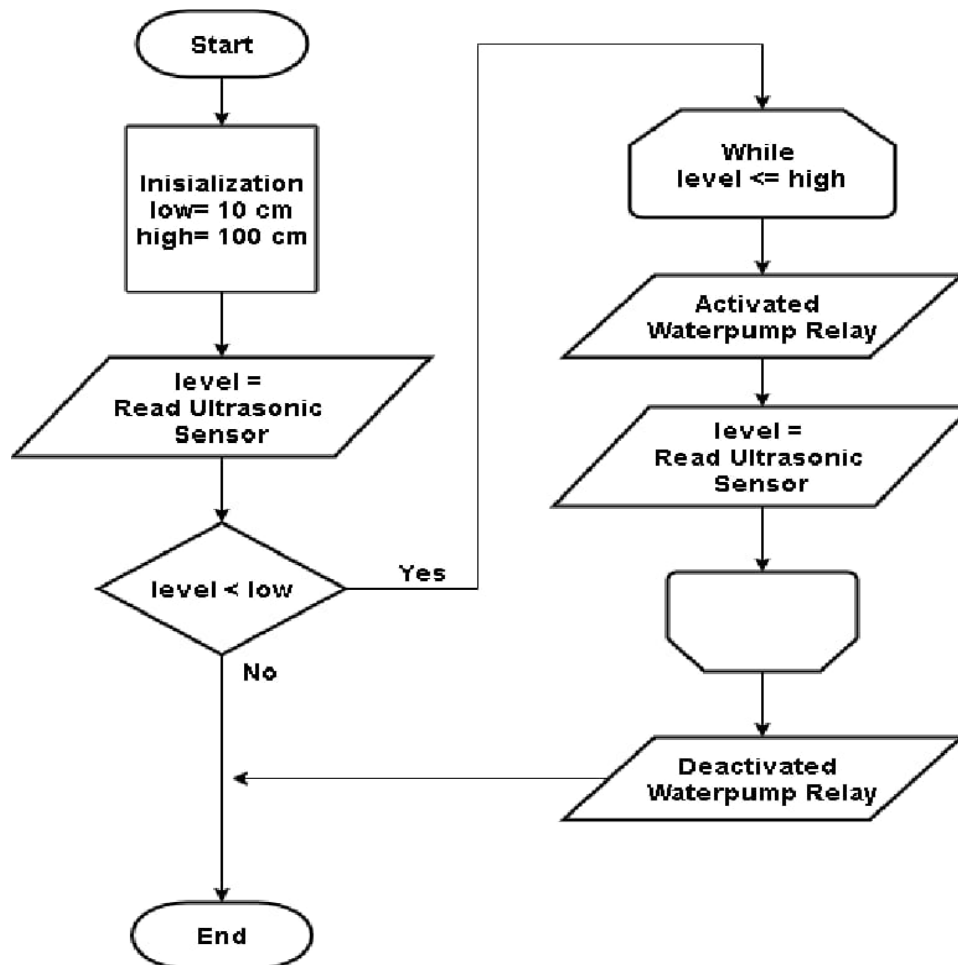


Figure 4.4: Use case diagram

4.2.3 Use case diagram

Use Case diagram is used to show the interaction between the different system.

Here, the User interacts with the system through:

- It acces the current state of the plant
- Consumers can manually turn extractors, wind turbines, and feeding pumps on or off, as necessary, whenever necessary.
- It is use to adjust the water ,light,moisture,temperature etc.

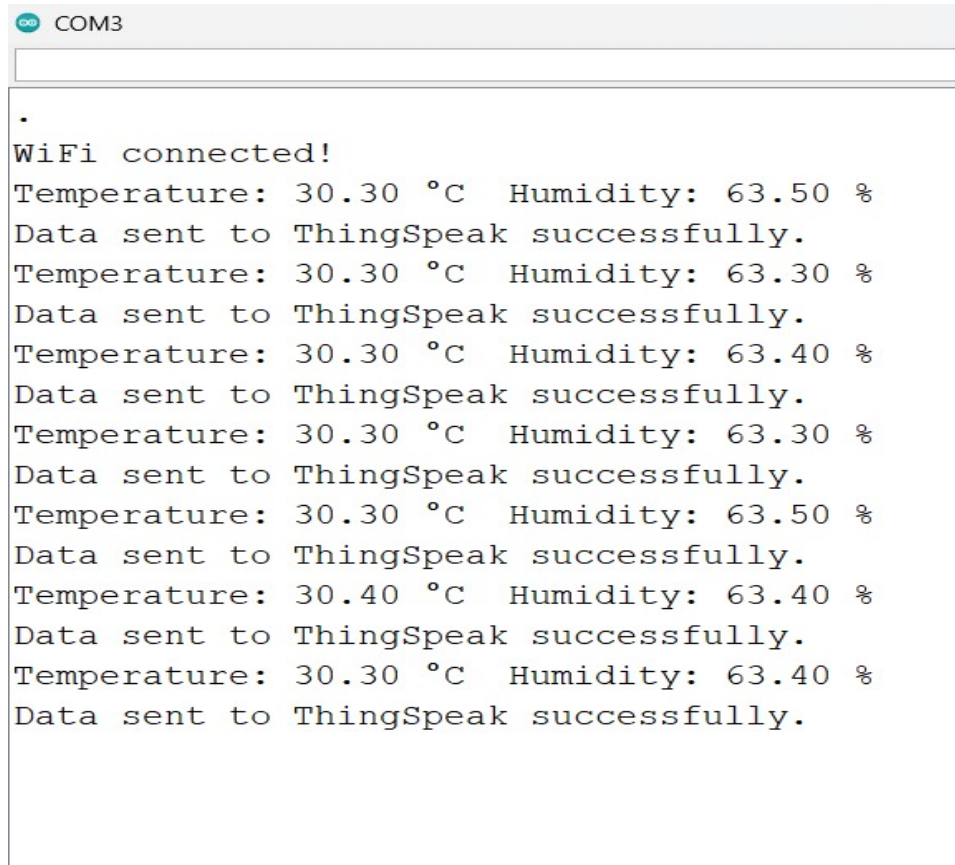
4.3 Control Flow Design

4.3.1 Activity diagram for use cases

An Activity Diagram represents the sequence of operations and workflow within the system. Despite this, assuming the item is wrong—such as reduced essential elements, increased temperature, or minimum water content—the diagram shows the following operation where the accountant triggers the relevant action, such as turning on the pump, turning on the fan, adding nutrients, or changing the light. Once the action has been completed, the framework checks the standards once more to confirm that the conditions have improved. Meanwhile, the entire modified reading is transmitted to the user interface and the cloud so that the user can see what is happening. This cycle continues to repeat itself, showing how the structure continuously protrudes, adjusts, and maintains the perfect ecosystem for plant growth.

4.4 Summary

A hydroponics operation diagram shows an endless cycle of how the framework work to look after the plant is performed. It starts, along with the detector that checks the surroundings and sends the reading to the accountant. The accountant analyses the facts and decides whether or not



```
COM3
.
WiFi connected!
Temperature: 30.30 °C Humidity: 63.50 %
Data sent to ThingSpeak successfully.
Temperature: 30.30 °C Humidity: 63.30 %
Data sent to ThingSpeak successfully.
Temperature: 30.30 °C Humidity: 63.40 %
Data sent to ThingSpeak successfully.
Temperature: 30.30 °C Humidity: 63.30 %
Data sent to ThingSpeak successfully.
Temperature: 30.30 °C Humidity: 63.50 %
Data sent to ThingSpeak successfully.
Temperature: 30.40 °C Humidity: 63.40 %
Data sent to ThingSpeak successfully.
Temperature: 30.30 °C Humidity: 63.40 %
Data sent to ThingSpeak successfully.
```

Figure 4.5: Activity diagram

everything is normal, assuming the need for an adjustment is necessary. Assuming that this is beyond the extent of the problem, the structure instinctively turns on the pump, light, admirers, or other nutrient delivery to solve the problem. After the correction, the framework checks the principles again to produce a certain state has been improved. Any additional information will be made available to the user via the App alternately displayed. This cycle continues to repeat itself, ensuring that plants will always grow in a solid and healthy environment.

Chapter 5

Implementation

5.1 Hardware Implementation

Applying hardware to a hydroponics organization, you can enjoy the construction of a smart gardener who appears next to your plant day and night. Finally, Wi-Fi or Bluetooth enables the system to transmit data to a mobile application or a splash screen, giving the customer complete control from anywhere. In general, hardware equipment brings together detectors, accountants, pumps, and network equipment to create an intelligent, self-regulating plant growing environment.

5.2 Software Implementation

Software implementation in hydroponics focuses on intelligent and spontaneous hardware creations. The software collects information from the entire detector, processes it, and decides when the pump, light, or other devotees must bend otherwise. It also defines threshold values, such as ideal pH or temperature ranges, so that an organization can correct itself at any time an object drifts out of balance. The software sends each modernized reading to a portable application, otherwise a splashboard, where the user can monitor the hardware in real time. In addition to storing information in the cloud, graphs, and gun trigger alerts, it takes into account " decreased water temperature " or " increased temperature ". In general, the software acts as the intelligent operator of the system, en-

suring the smooth functioning of the system and providing the customer with clear, simple supervision.

5.3 Summary

Complete software implementation for the creation of intelligent and robotic apparatus in the hydroponics industry. He reads detector information, decides what events the system should collect, and controls the pumps, lights, and fans to keep the plant in good order. In addition, a real-time reading of the customer, the shop information, and an alert is sent when the product is in need of attention. Generally speaking, software work benefits from the intelligent brains that ensure the smooth functioning of the hydroponic system.

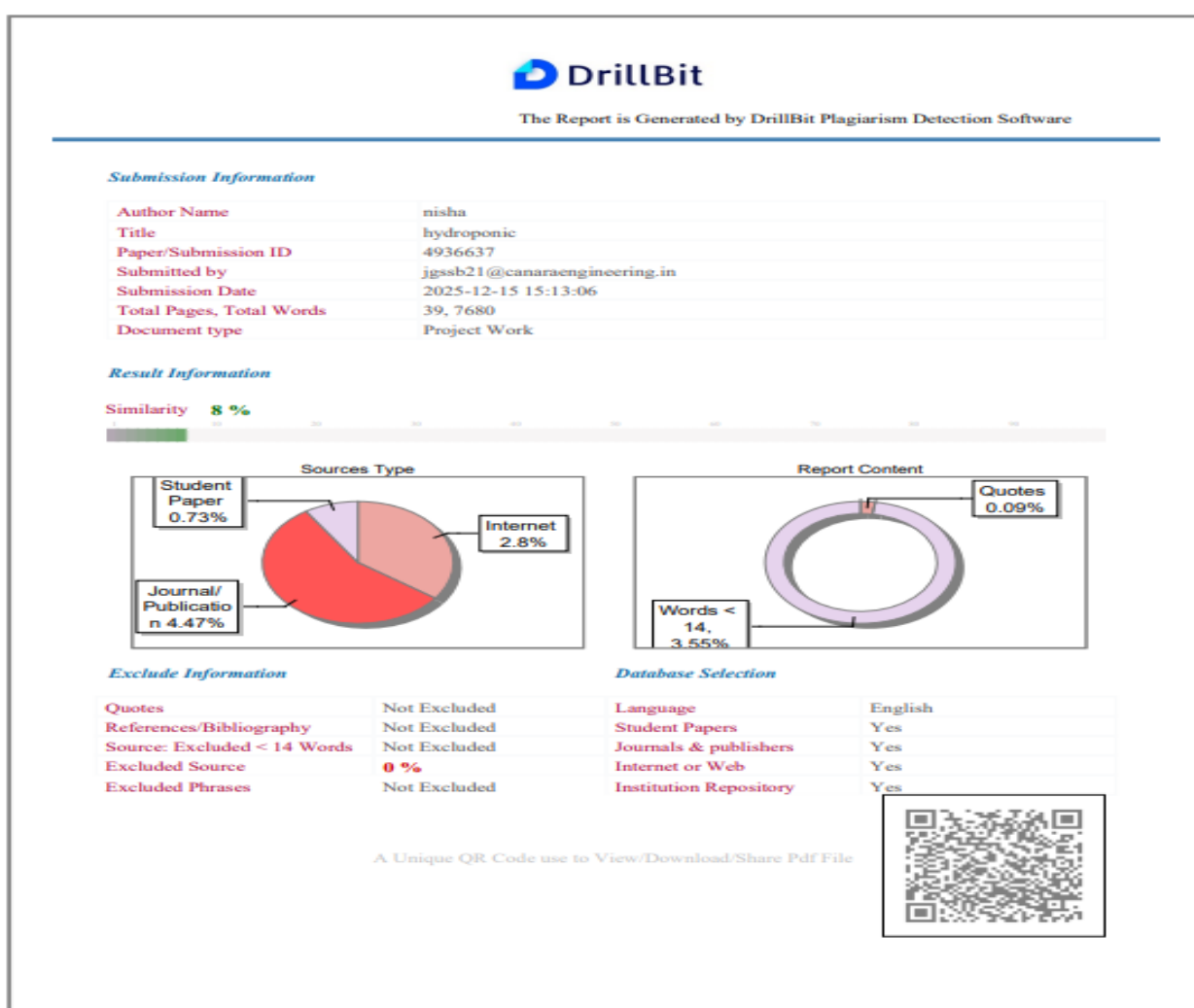
Chapter 6

Conclusions and Future Work

In the end, hydroponics architecture has succeeded in creating a smart and simplified mode for the development of plants free from dirt. In conclusion the hydroponics system is very much essential for growing the plants in nutrient rich water without any manual effort of the humans. the future work for these hydroponics is ,we can implement the solar batteries which will be very helpful to grow any types of the crops.

Appendix A

Drill-bit



Appendix B

Project-Expo Details

Certificate for participating in INNOVATION SHOWCASE project Exhibition 2025-2026



CANARA ENGINEERING COLLEGE
An Autonomous Institution, Approved by AICTE, Accredited by NAAC with A Grade
SUDHINDRA NAGARA, BENJANAPADAVU, BANTWAL, MANGALURU - 574219





CERTIFICATE
OF PARTICIPATION



Mr./Ms. Jitin Naik bearing USN 4CB23EC401 is hereby recognized for valuable participation in the Innovation Showcase (Project Exhibition – Exhibition Prototype) for the A.Y. 2025–26. Your involvement, effort, and enthusiasm contributed to the success of the event, and we proudly acknowledge your commitment and participation.

Title of Project: Hydroponic



Dr. Gurudeva Shastri Hiremath
Chief Coordinator
Major Project



Dr. Demian Antony D'Mello
Dean Academics - CEC



Dr. Nagesh H.R.
Principal